

What is a training example?

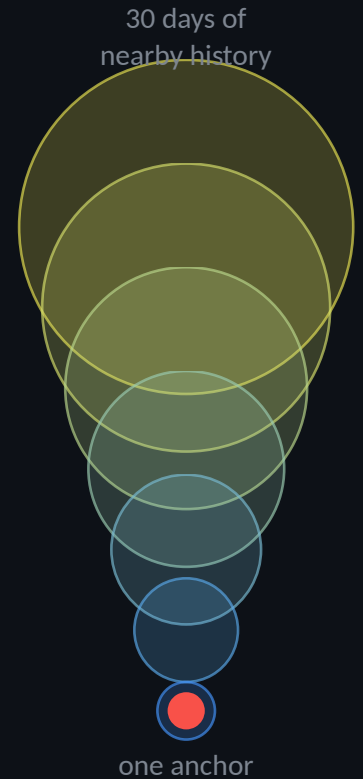
The cone codec's example, its boundaries, and where it should be materialized.

E-069 — the cone-native codec: an encoder that reads 30 days of nearby history around one ocean cell, so that motion can live inside the embedding.

2026-09-03

Proposal in one line

Keep generating examples on the fly — but make every example addressable, dumpable and certifiable.
Materialize the keys and the evaluation set, not the bytes.



One anchor, plus everything around it

A training example is **one anchor** — one ocean cell and one 5-day bin — together with the cone of history that surrounds it.

42

patch tokens

the 3×3 neighbourhood at lag 0, one per channel

706

dot tokens

single values at (channel, lag 1–6, offset) inside the inner cone

42 + 84

targets

the anchor's own values now, and at t+1 and t+2

4

context numbers

season (sin/cos of day-of-year), latitude, longitude

6,579 numbers ≈ 19 KB

Nothing is stored per example: it is gathered by index from the 36 GB tensor at batch time.

750 tokens enter attention (two bookkeeping tokens + 42 + 706) · a batch of 256 examples ≈ 6 MB

Seven bins of history, two bins of future



reach grows with lag — $129.6 \times (1+\text{lag})$ km for currents and sea-surface height

A wind stress, storminess

8 dots · 500 km at lags 0-1, then nothing

B currents, sea-surface height, Argo column

80 dots · disc grows $129.6 \times (1+\text{lag})$ km

C sea-surface temperature, mixed layer

81 dots · 500 → 388.8 → ... → 907 km

rg Argo depth channels

6 dots · anchor column only, no disc

Dependency cone (E-069)

388.8 km
family B · inner cone

123
dots per channel

1036
tokens per anchor · 42
patch + 994 dots

what-if geometry — not what the codec was trained on

family B · currents & SSH

lag sweep lags

lag 2 · 10 days back · 2017-05-27 held-out year
anchor 36.00° N 70.00° W · 15 dots drawn

date 06/06/2017

☐ 0.5 s / pentad

☐ follow the cone — drive the globe's date

☒ trace

live on the app's Cones tab

The dataset is a distribution, not a list of files

thing	size	when made	keyed by
the input tensor (family 4 r3)	35.7 GB float16	once, sha-pinned (fa460837fa)	its sha256
the anomaly-space scratch copy	36 GB	per run, ~40 min	(per run)
per-row dot tables (281 rows)	tiny	per run, 0.8 s	geometry constants
the observed mask	computed per gather	—	—
the fixed evaluation anchor set	1,024 anchors	per run, from seed+991	seed
the velocity-probe anchors	2,048	per run, seed+7717	seed
the training anchor stream and mask draws	implied	per run, from the seed	seed
(stage 2) the embedding cache Z	17.4 GB float16	once per codec × tensor	codec-weight hash × tensor sha

Everything else — the example itself — is derived on the fly.

Uniform draws from 251 million anchors

2,891

of 3,142 bins are admissible

all nine bins $t-6 \dots t+2$ must exist and be training bins

$\approx 86,700$

ocean cells

any cell where a current was ever observed

$\approx 251 \text{ M}$

anchors in the pool

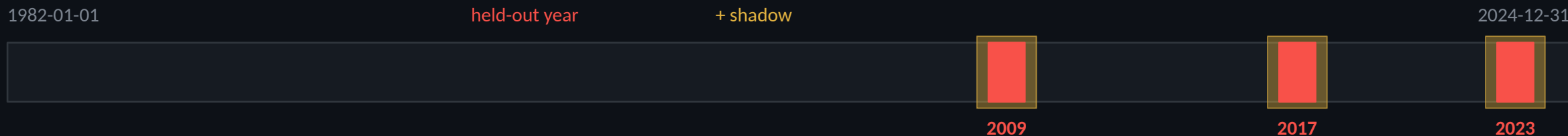
drawn uniformly, with replacement — there are no epochs

What a $20,000 \times 256$ run actually sees

5.1 million anchors = 2 % of the pool

- 251 bins excluded: the three held-out years 2009 / 2017 / 2023 with their 6-bin shadows, and the axis ends.
- No duplication in storage, and near-none in sampling — but every tensor value is read by roughly 700 different examples.
- That 700× is exactly the duplication a materialized dataset would have to pay in bytes.

The held-out years, and the shadow around them



Zoom — one year edge, in 5-day bins



Training — certified

The whole cone ($t-6 \dots t+2$) must avoid held-out bins. Enforced twice: by the admissibility rule, and by an independent brute-force certificate over 4,096 drawn anchors before the model is built — 0 violations, or the run refuses to start. Runs #536 and #537 both printed 0.

Evaluation — not certified

Only the anchor bin must be held out; the cone's context may reach into training bins. About 11 % of evaluation anchors sit within 6 bins of a year edge and straddle. Deliberate — “a held-out measurement, not a second training pool” — but an asymmetry.

The axis ends: cones that would run off 1982-01-01 or past 2024-12-31 are refused from every pool, never padded.

The edge of the map, land, and the dateline

0-70°N



○ off-window dot — kept as a token, valid = false, value 0, and never a target

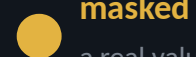
The global family (E-070) will wrap at the dateline — a 10-line change, planned.

Three kinds of token



real

a measured value the model may read



masked

a real value we hid on purpose — and ask for back



missing

land, never observed, or off the window — never a target

Existence is enforced in exactly one place: one key-padding mask. A test flips an invalid dot's value and asserts the embedding is bit-identical.

Observation gaps are represented, never imputed

≈ 83 %

of depth-channel dots carry the missing token — even inside the Argo era

Argo is live one bin per month — one pentad in six



five months of 5-day bins — filled bins are the ones Argo actually reports

22 of 43 years carry no live Argo bins at all.

- Missing is never forward-filled: telling the model the subsurface was observed on a day it was not would be a lie in the data.
- An unobserved target has weight exactly 0 — the decoder is never scored against a made-up zero.
- The token shape stays fixed (706 dots at every latitude); liveness rides in a flag, not in the shape.

Missingness is information — so it is a token, not a hole.

What the example is actually used for

Hide

- **whole channels** — currents 50 %, others 30 %
- **a band of recent lags** — 30 % of examples, lags 1 ... l_0
- **a 90° sector** — 30 % of examples

Score

Gaussian negative log-likelihood, weighted by family (A 1.0, B 2.0, C 1.5). The headline term reads the 32-number embedding z alone.

Ask 382 questions

- 42** reconstruct the anchor's own values
- 84** predict the future values at $t+1$ and $t+2$
- 256** recover hidden dots from elsewhere in the cone

The snapshot twin ($L_{in} = 0$)

44 tokens, no history, still asked for $t+1$ and $t+2$ — the H1 control: does the embedding carry velocity?

The forecaster's example is a different object

A sequence of embeddings, not raw values

- Reads 144 lags (2 years) of embeddings at the anchor column, plus the outer spiral — 24 cells per lag from lag 7.
- Embeddings come from the materialized Z cache (17.4 GB), gathered on the fly per window.
- Off-window cells are zero-filled; windows must not touch any held-out bin, including each frame's teacher-forced target.
- That rule was derived once and then re-counted by brute force: 2,779 → 2,417 usable end-bins.

3,432

cells read per window — the new annulus stencil

20,880

cells read by the older full cylinder

Status: the annulus stencil is the one E-069 piece not yet written.

Generate on the fly, or materialize every example?

Option 1 — generate on the fly (today)

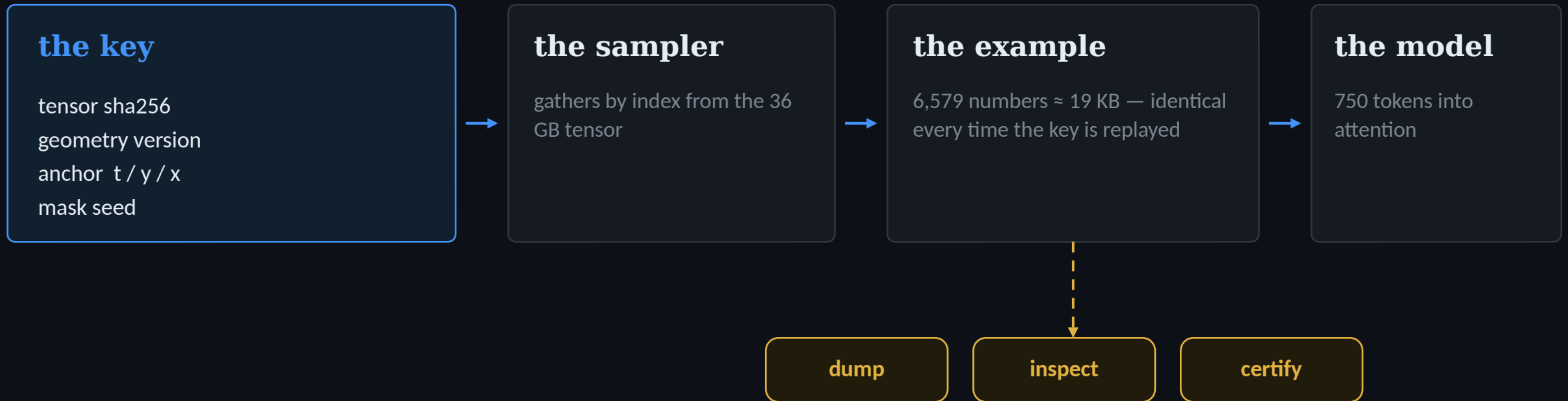
Option 2 — materialize each example

Storage	0 per example	251 M × 19 KB ≈ 4.8 TB for the pool; one run’s stream ≈ 97 GB
Duplication	none	700× — each cell sits inside ~700 cones
What an example is	a pure function of (tensor sha, geometry, anchor, mask seed)	bytes on disk, with boundary flags baked in
Boundary handling	the sampler + a certificate that runs every dispatch	the pipeline — but a sampler bug becomes permanent data
Changing the geometry	costs nothing	rebuilds everything
Inspectability	not inspectable without running code	examples are files you can open; sequential reads are fast

Runtime cost of option 1: 0.2–0.8 ms per anchor of host gather — about 1 hour of a 20,000-step run, not overlapped with the GPU today.

Materialize the key, not the bytes

An example is fully determined by a **small key**. Materializing the bytes is caching; materializing the key gives the same inspectability at zero storage — provided a tool turns a key back into the example, bit-identically.



The repo already works this way: the stage-2 cache is keyed by two hashes, and the JAX port is certified by gates rather than by re-reading weights.

What I would do — about one day of work

1 A dump-by-key tool

Dump any example by key — or by “run #N, step s, batch index b”, replaying the seeded stream — to NPZ, JSON and a picture. The Cones tab gains a “show a real example” mode.

2 A run manifest

Tensor sha, geometry json sha, seed and step count written into every run’s record, so any example ever trained on is regenerable.

3 Materialize the evaluation set

The 1,024 evaluation anchors and their fixed mask draws, as a versioned file shipped with each run. This is where materialization pays.

4 An executable boundary spec

Property tests sweeping every edge row and column, both axis ends and every holdout edge; the certificate extended to evaluation anchors behind `--eval-scope {anchor, window}`.

5 Materialize the observed mask

Once per tensor — a 2.2 GB bitmask — and precompute the row tables. Removes the double read and part of the 0.2–0.8 ms.

6 An escape hatch

`--materialize-stream` writes a run’s exact examples (≈ 97 GB) for audits. Never for training.

Cost: about one day of implementation • no change to E-069’s numbers • the evaluation file becomes an artefact.

Boundary handling under each option

BOUNDARY	ENFORCED TODAY	UNDER OPTION 2	HOW IT IS PROVEN
time / held-out years	the admissibility rule in the sampler	a filter at build time	certificate over 4,096 anchors, before every run
window edge	dot kept, valid = false; one key-padding mask	a flag baked into each stored example	test: flip an invalid dot, embedding is bit-identical
land / never observed	the missing token, at gather time	a missing flag written once at build	by construction, plus gather tests
dateline / wrap	no wrap; E-070 will wrap — a 10-line change	a full rebuild of every byte	geometry test (planned)
axis ends 1982 / 2024	cones refused from every pool, never padded	rows simply absent from the built set	admissibility count + the same certificate

Under option 1 the proof is a test that runs every dispatch. Under option 2 the proof is a build that ran once.

Four questions for you

Generation

Keep on-the-fly generation, as proposed — or materialize a run's stream by default?

Evaluation scope

Keep “anchor bin held out, context may straddle” — or measure the strict variant too?

A public gallery

A small gallery of real examples — say 50, about 1 MB — for the Cones tab and the paper. Yes or no?

Order of work

After run #537's H1 reading, or in parallel? It touches no training code path except the evaluation-set file.

Recommendation: parallel, small, and the strict evaluation variant measured once — so the straddle's effect is a number rather than an assumption.